

Grade 7 Accelerated  
Unit 3  
Lesson 9 Practice

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

1. What is the value of  $10^{-16}$ , written as a fraction?

2. Complete the statements.

The expression  $\frac{10^{-17}}{10^{-5}}$  can be written as \_\_\_\_\_. Another way to write the expression  $\frac{10^{-17}}{10^{-5}}$  using positive exponents only is \_\_\_\_\_.

3. Complete the table. Write each mathematical statement in an equivalent form by applying one or more laws of exponents. Then, write the equivalent decimal value.

| Mathematical Statement                | Value Written as a Power of 10 | Decimal Equivalent |
|---------------------------------------|--------------------------------|--------------------|
| $(10^4) - 4 \times 10^{25}$           |                                |                    |
| $(\frac{1}{10^{-5}})^3$               |                                |                    |
| $\frac{10^{-14}}{10^{-9}} \cdot 10^6$ |                                |                    |

Express each number in scientific notation.

4. The diameter of a human hair is about 67 hundred-thousandths of an inch.

5. 0.0000000049

Express each number in standard form.

6. A cell's diameter  $0.0174 \times 10^{-4}$

7.  $9 \times 10^{-6}$

8. Complete the table.

| Scientific Notation  | Standard Form |
|----------------------|---------------|
|                      | 0.000000026   |
| $5.2 \times 10^{-3}$ |               |

9. Complete the statements using the table in question 8.

○ 5.2 is \_\_\_\_ times 

○ smaller

○ larger

 than \_\_\_\_.

○  $10^{-3}$  is \_\_\_\_times 

○ smaller

○ larger

 than  $10^{-4}$ .

○ The value  $5.2 \times 10^{-3}$  is \_\_\_\_ times 

○ smaller

○ larger

 than 0.000000026.